

Skills Summary

Sums of Radicals

Use this reference while completing your worksheet or when studying.

Skill 1 — Testing whether two roots may be added

The rule that is broken most often: a radical does *not* distribute across addition.

$$\sqrt{a} + \sqrt{b} \neq \sqrt{a+b}$$

How to test it: pick numbers whose roots you know, work out both sides, and compare. One counterexample settles the question for good.

Example: Is $\sqrt{4} + \sqrt{9}$ the same as $\sqrt{4+9}$?

Step 1. The claim is a general one, so a single pair of numbers with known roots is enough to decide it — pick perfect squares and compute both sides.

Step 2. The left side is two whole roots added; the right side is one root of the total, which is not a perfect square and lands near three and a half.

Left side = 5, right side = $\sqrt{13} \approx 3.61$. Not equal.

Try it: Work out $\sqrt{25} + \sqrt{144}$, then $\sqrt{169}$.

check: 17 and 12

Skill 2 — Combining like radical terms

Rule: treat the radical as a name. Terms with the *same* radical part are like terms, so add or subtract only the counts in front:

$$p\sqrt{k} + q\sqrt{k} = (p+q)\sqrt{k}$$

The radical part never changes when like terms are collected. Terms with different radical parts simply do not combine.

Example: Simplify $6\sqrt{7} + 2\sqrt{7}$.

Step 1. Both terms carry the same radical part, so they are like terms and only the counts in front are added.

Step 2. Counting: six of them plus two more of them makes eight of them, and the radical part is carried along unchanged.

$$\Rightarrow 6\sqrt{7} + 2\sqrt{7} = 8\sqrt{7}$$

Try it: Simplify $9\sqrt{5} - 4\sqrt{5}$.

check: $5\sqrt{5}$

(!) **Watch out:** the two classic slips are adding the numbers *under* the radicals, and doing that on top of adding the counts in front. Check every answer by asking: is the radical part exactly what it was before?

Skill 3 — Simplify first, then combine

Method: radical parts that look different can become the same after simplifying. Pull the largest perfect-square factor out of each radicand, then collect like terms.

$$\sqrt{p \cdot k} = \sqrt{p} \cdot \sqrt{k} \quad \text{when } p \text{ is a perfect square}$$

Example: Simplify $\sqrt{12} + \sqrt{27}$.

Step 1. The two radical parts differ, so nothing can be combined yet — pull out perfect-square factors first and see whether they match.

Step 2. Each radicand is a perfect square times three, so each term becomes a whole-number count of the same radical, and the two counts then add.

$$\Rightarrow \sqrt{12} + \sqrt{27} = 5\sqrt{3}$$

Try it: Simplify $\sqrt{20} + \sqrt{45}$.

check: $5\sqrt{5}$